DIGITAL FORENSICS INVESTIGATION REPORT

BlueSky Ransomware Lab – Network & Event Log Forensics

Case Name	BlueSky Ransomware Lab
Case Number	2026-012
Examiner	Tej Chaute
Date	08/04/2026
Platform	CyberDefenders
Evidence File	BlueSkyRansomware.pcap, BlueSkyRansomware.evtx
SHA256 Hash	A382838772DDC80FA9824676F2FC55D5907FF79424DD0EDCCF15CF19DE7CB6BB, 5E501D3B0E97563975362AECD5C64911E4432C62E6327C8AB2067DCBF0C59309
Classification	Training

1. Case Summary

A suspected ransomware incident was investigated by analysing captured network traffic and Windows Event Log artifacts. The investigation revealed a full attack lifecycle spanning two dates: an initial PowerShell execution under winlogon.exe was recorded in event logs on 2024-04-23, indicating an earlier access or staging phase, while the primary attack activity — MSSQL exploitation, payload delivery, credential dumping, and ransomware deployment — occurred on 2024-04-28.

The attack exploited an exposed MSSQL service using the sa account, leveraged xp_cmdshell for OS command execution, delivered multiple PowerShell payloads to disable defences, dump credentials, and establish persistence, and culminated in the deployment of the BlueSky ransomware binary (javaw.exe), resulting in file encryption and ransom note creation. Attribution to the BlueSky ransomware family was confirmed via VirusTotal sandbox analysis.

2. Evidence Examined

Item	Filename	Hash (SHA256)	File Size
001	BlueSkyRansomware.pcap	A382838772DDC80FA9824676F2FC55D5907FF79424DD0EDCCF15CF19DE7CB6BB	2.10 MB
002	BlueSkyRansomware.evtx	5E501D3B0E97563975362AECD5C64911E4432C62E6327C8AB2067DCBF0C59309	1.06 MB

3. Tools Used

Tool	Version	Purpose
Wireshark	4.6.4	Network traffic analysis — TDS/MSSQL inspection, HTTP stream reconstruction, payload extraction
CyberChef	10.23.0	Base64 decoding of obfuscated PowerShell payloads
Event Viewer	Windows	Windows Event Log analysis — PowerShell execution, process context
VirusTotal	2.5.8	Malware classification, ransom note identification, sandbox behavioural analysis

4. Key Findings

Each finding below is supported by at least one artifact. Path, timestamp, and source tool are documented for every item.

4.1 System Information

Field	Value
Computer name	DESKTOP-7EQVM78

Field	Value
Timezone	UTC
Attacker IP	87.96.21.84
Victim IP	87.96.21.81
Evidence Note	OS version and architecture not recoverable from available PCAP and EVTX artifacts

4.2 Artifact Table

#	Artifact Type	Location / Path	Timestamp (UTC)	What It Proves
1	Network Traffic (Port Scan)	PCAP (TCP SYN scan) 87.96.21.84 → 87.96.21.81	2024-04-28 00:29:56	Attacker performed port scan against the target, identifying exposed services — confirms start of deliberate reconnaissance phase prior to exploitation
2	MSSQL Authentication	PCAP (TDS Login)	2024-04-28 00:30:13	Attacker authenticated to the MSSQL service using the sa (system administrator) account — confirms credential-based initial access to the database service
3	SQL Command Execution	PCAP (TDS Query) xp_cmdshell enabled	2024-04-28 00:30:13	xp_cmdshell enabled via SQL query — grants the attacker OS-level command execution through the database engine, escalating from database access to system command execution
4	PowerShell Execution (winlogon.exe)	Windows Event Log (BlueSkyRansomware.evtx) Event ID 400 — Engine Lifecycle HostApplication: winlogon.exe HostName: MSFConsole	2024-04-23 15:31:18 (5 days prior to PCAP activity)	PowerShell engine initialised inside winlogon.exe — HostApplication=winlogon.exe in Event ID 400 confirms PowerShell was hosted within the Windows logon process, indicating process injection or SYSTEM-context execution. HostName=MSFConsole confirms Metasploit Framework was the originating console. Timestamp predates PCAP by

#	Artifact Type	Location / Path	Timestamp (UTC)	What It Proves
				five days, indicating an earlier access or staging phase
5	Payload Download (checking.ps1)	PCAP (HTTP GET)	2024-04-28 00:32:10	First PowerShell payload downloaded from attacker-controlled server — initiates the post-exploitation scripted phase of the attack
6	Defense Evasion	PowerShell Script	2024-04-28 00:32:10	Script disables Windows Defender protections, allowing attacker activity without detection.
7	Persistence Mechanism	Scheduled Task	2024-04-28 00:32:10	**del.ps1** creates scheduled task \Microsoft\Windows\MUI\LPupdate to maintain persistent execution of the malicious payload — task name masquerades as a legitimate Windows MUI language pack update to evade detection
8	Credential Dumping Tool	PCAP (HTTP GET) Invoke-PowerDump.ps1	2024-04-28 00:32:12	Credential dumping script retrieved from attacker server — confirms intent to extract NTLM password hashes from the system for potential lateral movement or offline cracking
9	Credential Storage	Decoded Payload	2024-04-28 00:32:12	Extracted credential hashes written to hashes.txt — confirms successful credential harvesting from the compromised system
10	Reconnaissance Data	PCAP / decoded script extracted_hosts.txt	2024-04-28 00:32:12	Host enumeration data compiled into extracted_hosts.txt — attacker built a list of internal network targets, confirming lateral movement preparation
11	Ransomware Payload	PCAP (HTTP GET) javaw.exe	2024-04-28 00:32:21	Ransomware binary downloaded from attacker server — final-stage payload delivery confirming the attack culminates in ransomware deployment

#	Artifact Type	Location / Path	Timestamp (UTC)	What It Proves
12	Ransom Note	VirusTotal (sandbox behavioural analysis)	—	File '# DECRYPT FILES BLUESKY #.txt' identified in VirusTotal sandbox output — confirms successful ransomware execution and file encryption, with ransom note dropped to victim system
13	Malware Attribution	VirusTotal (SHA256: javaw.exe hash)	—	Hash correlation via VirusTotal identifies javaw.exe as BlueSky ransomware family — enables threat intelligence correlation and confirms the specific ransomware variant deployed

5. Event Timeline

Reconstructed chronological sequence of events based on artifact timestamps. All times in UTC. Note: the event log entry from 2024-04-23 predates PCAP activity by five days, indicating an earlier access phase that falls outside the network capture window.

Timestamp (UTC)	Artifact Source	Event Description
2024-04-23 15:31:18	Windows Event Log (BlueSkyRansomware.evtx) Event ID 400	PowerShell engine initialised inside winlogon.exe (HostName: MSFConsole) — confirms SYSTEM-context PowerShell execution via Metasploit, five days before PCAP activity. Indicates an earlier staging or access phase not captured in network traffic
2024-04-28 00:29:56	PCAP	Attacker (87.96.21.84) performed port scan against target (87.96.21.81) — reconnaissance to identify exposed services including MSSQL
2024-04-28 00:30:13	PCAP	MSSQL login using sa account observed — credential-based initial access to database service confirmed
2024-04-28 00:30:13	PCAP	xp_cmdshell enabled via SQL query — OS-level command execution gained through the database engine

Timestamp (UTC)	Artifact Source	Event Description
2024-04-28 00:32:10	PCAP	Victim downloads checking.ps1, initiating attacker-controlled PowerShell execution.
2024-04-28 00:32:10	Script Analysis	PowerShell payload disables Windows Defender and modifies security settings.
2024-04-28 00:32:12	PCAP	del.ps1 downloaded — script created scheduled task \Microsoft\Windows\MUI\LPupdate for persistence
2024-04-28 00:32:12	PCAP	Invoke-PowerDump.ps1 retrieved — credential hash dumping script executed, hashes stored in hashes.txt
2024-04-28 00:32:12	Script Analysis	extracted_hosts.txt retrieved — internal host enumeration data compiled for lateral movement preparation
2024-04-28 00:32:21	PCAP	javaw.exe (BlueSky ransomware) downloaded from attacker server — final-stage payload delivered, file encryption and ransom note deployment follows

6. Conclusion

The investigation confirms the system DESKTOP-7EQVM78 was compromised through a structured, multi-stage ransomware attack with activity spanning two dates.

Early Access (2024-04-23): Event log analysis revealed PowerShell execution inside winlogon.exe with HostName=MSFConsole five days before the primary attack, indicating an earlier Metasploit-based access or staging phase not captured in the network traffic. This suggests the attacker had prior presence on the system before the PCAP recording began.

Initial Access and Execution (2024-04-28): The attacker exploited the MSSQL service using the sa account (Artifact #2) and enabled xp_cmdshell for OS-level command execution (Artifact #3). The first PowerShell payload (checking.ps1) was downloaded and decoded to reveal Windows Defender disabling commands (Artifacts #5–6), removing security coverage before further payloads were delivered.

Persistence and Credential Harvesting: del.ps1 established persistence via a scheduled task named \Microsoft\Windows\MUI\LPupdate, masquerading as a Windows language pack update (Artifact #7). Invoke-PowerDump.ps1 was retrieved and executed to dump NTLM credential hashes to hashes.txt (Artifacts #8–9), with internal host enumeration compiled in extracted_hosts.txt for lateral movement preparation (Artifact #10).

Ransomware Deployment: javaw.exe was downloaded as the final-stage payload (Artifact #11). VirusTotal sandbox analysis confirmed the binary as BlueSky ransomware (Artifact #13), which encrypted files and dropped the ransom note '# DECRYPT FILES BLUESKY #.txt' (Artifact #12). The full attack chain — MSSQL exploitation, PowerShell staging, defence evasion, credential theft, persistence, and ransomware deployment — represents a complete multi-stage compromise lifecycle.

7. Examiner Notes

This investigation was conducted retrospectively as part of a structured DFIR learning path. It is the first case in this portfolio combining network capture and Windows Event Log analysis for a ransomware incident.

Key methodological observations from this case:

- Process injection was not visible in the PCAP — the event log was the only source confirming PowerShell execution inside winlogon.exe. This reinforced the importance of always analysing host-based artifacts alongside network captures; relying solely on PCAP would have missed the most significant privilege escalation indicator in the investigation.
- The five-day gap between the event log timestamp (2024-04-23) and the PCAP activity (2024-04-28) is a significant investigative finding. The HostName=MSFConsole field in Event ID 400 confirms this was Metasploit-driven, suggesting the attacker had established SYSTEM-level access well before the ransomware deployment phase. In a real investigation this gap would require further analysis — additional event logs, authentication records, or endpoint telemetry — to determine what occurred between the two dates.
- Base64 decoding of the PowerShell payloads via CyberChef was essential for understanding attacker intent. checking.ps1 appeared benign from the HTTP request alone — the actual content (Defender disabling commands) was only revealed after decoding. Similarly, del.ps1's persistence mechanism was only identified through decoding, not from the filename or network metadata.
- The scheduled task name \Microsoft\Windows\MUI\LPupdate is a deliberate masquerade as a Windows language pack update task. Analysts reviewing scheduled tasks quickly would likely overlook it. Verifying the legitimacy of scheduled task names — particularly those under \Microsoft\Windows\ paths — by checking the associated binary path is an important habit that this case reinforces.

- VirusTotal was used for both malware attribution (javaw.exe hash) and ransom note identification. Local string extraction and network analysis alone could not confirm the malware family — pivoting to threat intelligence was necessary to close the attribution gap. This mirrors the Lockdown lab experience, where DNS and string analysis produced noise and VirusTotal provided the definitive answer.
- OS version could not be reliably determined from the available PCAP and EVTX artifacts. In a real investigation, the Security event log (Event ID 4624 or 4688) or SYSTEM registry hive would typically provide this. The absence of OS version is documented in the System Information table rather than omitted silently.

If this were a real-world investigation, recommended next steps would include: checking for lateral movement across hosts listed in extracted_hosts.txt; searching for available BlueSky decryptors; conducting memory forensics for deeper credential and process analysis; recovering the full contents of hashes.txt for credential rotation; and performing a broader hunt for Metasploit persistence mechanisms given the confirmed MSFConsole access five days before the ransomware deployment.